ASSESSMENT OF THEMATIC MAPPING TECHNIQUES FOR POPULATION DISTRIBUTION VISUALIZATION: A Comparative study of Dot Density, Filled Isopleth and Unfilled Isopleth

Abstract

Thematic maps play a critical role in communicating spatial patterns and supporting geographic decision making. Among the various methods used to represent population density, Dot Density maps, Filled Isopleths (Colour-graded), and Unfilled Isopleths (Contour-based) are widely applied, yet their influence on users' perception and interpretation of spatial density patterns remains insufficiently understood. This study investigates how these three cartographic representations affect the perception of population density, with particular emphasis on hotspot identification and spatial gradient interpretation. A between-subjects online experiment was conducted using maps representing both real-world and synthetic population distributions. Participants completed tasks while measure of accuracy, response time, and perceived difficulty were collected. Demographic information and self-reported cartographic experience were also recorded.

The findings demonstrate that thematic map design substantially affects the interpretation of population density patterns. Filled Isopleths appear to provide the most effective support for both hotspot identification and gradient interpretation, while Dot Density maps remain useful for rapid exploratory assessment of spatial clustering. These results contribute to cartographic perception research and provide practical recommendations for selecting thematic map types in population mapping and related spatial analysis applications.

Keywords: Cartographic perception, Thematic Maps, Population Density, Filled isopleth, Unfilled Isopleth, Dot Density, Real-World, Synthetic, Hotspot Identification, Spatial Gradients, Map Usability, Spatial Cognition.

1.Introduction

Thematic maps are special types of maps that show information about a topic, such as population, income, or temperature. They are important in cartography to visualize spatial patterns and support analysis (Slocum et al., 2023).

The visualization of spatial phenomena plays a critical role in understanding geographic patterns and support decision making processes. Thematic maps are widely used to communicate complex spatial information effectively through appropriate symbolization and map design (Slocum et al., 2023). In the context of natural hazard assessment, thematic mapping techniques can be used to represent population distribution built-up areas, and hazard exposure. Ehrlich et al., (2018) demonstrated the importance of integrating population density and built-up area datasets derived from remote sensing to quantify exposure to natural hazards to multiple spatial scales. Several mapping techniques are commonly used to represent population distribution. Dot Density maps use repeated dots to represent quantities within geographic space, enabling users to visually identify clustering, concentration and dispersion patterns. Filled Isopleth represent continuous spatial variation through graduated colour surfaces or heatmap-style visualization, allowing smooth transitions between density values. In contrast, Unfilled Isopleth, commonly referred to as isolines or contour maps, use lines connecting points of equal value to represent spatial gradients and continuous phenomena without filled colour regions. Unlike discrete spatial representations, isopleth-based visualization estimates values continuously across geographic space (Traun et al., 2017; He et al., 2011; Slocum et al., 2023). All three techniques are valid; each emphasizes different spatial characteristics and may lead to distinct interpretations by map readers. Previous studies have shown that thematic map design can influence user effectiveness, attractiveness ratings, and overall evaluation of spatial information. Furthermore, different visualization methods may affect how spatial patterns are perceived and interpreted, highlighting the importance of selecting an appropriate mapping technique for a given task. Selecting the most appropriate method therefore plays a crucial role in thematic design (Medynska-Gulij et al., 2020; Besançon et al., 2020; Forrest & Medynska-Gulij, 2021).

The effectiveness of these thematic mapping techniques is closely related to cartographic communication and spatial cognition, as different visualization methods requires user to interpret spatial information through varying levels of abstraction and visual complexity (MacEachren 2004). However, different thematic mapping techniques emphasize different characteristics of spatial data. Because these visualization methods communicate information in different ways, they may influence how users perceive, interpret, and evaluate spatial patterns.

1.1 Map Interpretation and User-Centred Design

Research has shown that the way information is represented on a map influences not only what users see but also how they understand and interpret spatial relationships (MachEachren, 2004). Thematic maps communicate spatial information through abstraction, classification, and symbolization, enabling complex geographic phenomena to be represented in a form that facilitates pattern recognition and interpretation (Ding & Meng, 2014). Modern cartography therefore adopts a user-centred perspective, recognising that perceptual and cognitive factors play an important role in map use and interpretation (Roth et al., 2017).

Recent studies have demonstrated that users do not always interpret maps in the same way. Map-reading performance is influenced not only by the visual characteristics of a map but also by the characteristics of the user and the task being performed. Cognitive and usability studies have shown that different map designs can affect the accuracy and efficiency with which users extract and interpret spatial information (Pickle, 2004). Furthermore, no single map design is optimal for all users or analytical task, highlighting the importance of evaluating cartographic visualizations from a user-centred perspective (Pickle, 2004). Similarly, Medynska-Guly et al. (2023) showed that user testing offers important insights into how well thematic maps work in practice, highlighting the gap between what users like and what helps them perform better.

The findings of Beitlova et al. (2020) further demonstrate that cartographic design can significantly affect map comprehension and task success. Their study showed that users experienced varying levels of difficulty across thematic map types, with performance influenced by factors such as symbol interpretation, legend use, and visual complexity. The authors emphasized that effective map design should support users in identifying patterns, locating information, and completing analytical tasks accurately and efficiently. Consequently, evaluating different thematic mapping techniques through measures such as accuracy, response time, and user perception is essential for understanding their effectiveness in communicating spatial information.

These studies collectively suggest that it's important to carefully manage how much visual information and complexity a thematic map contains. A good map should present enough detail to be useful while remaining clear and easy to understand. This helps people to complete tasks like finding locations, making comparisons, or spotting patterns without getting confused. By testing and comparing different type of maps, researchers can figure out which designs provide the best mix of accuracy and user satisfaction.

1.2 Research Gap

While several studies have examined thematic mapping principles and cartographic visualization techniques, relatively few have conducted comparative user-based evaluations involving Dot Density, Filled Isopleth (colour graded) and Unfilled Isopleth (contour lines) maps within the same experimental framework. Recent research has demonstrated that the choice of thematic map type can significantly influence map-reading performance and spatial interpretation. For instance, Sukraini et al. (2022) investigated the effectiveness of choropleth and graduated symbol maps for communicating COVID-19 case distributions in Bali, Indonesia. A total of 120 participants completed eleven map-reading tasks, including identifying, comparing, interpreting, categorizing, and locating spatial information. The study evaluated user performance based on three metrics: accuracy response time, and level of difficulty. Results showed that participants using choropleth maps achieved higher accuracy (73%) compared with those using graduated symbol maps (63%). In addition, choropleth maps resulted in faster task completion times, with an average response time of 28.15 seconds and compared to 35.78 seconds for graduated symbol maps. Participants also perceived choropleth maps as easier to read and understand. Based on these findings, the authors concluded that choropleths maps provided more effective means of communicating spatial patterns and supporting map-reading tasks than graduated symbol maps. Similar findings were reported by Słomska-Przech and Gołębiewska (2021), who compared choropleth, graduated symbol, and isoline maps using a large-scale user study involving 366 participants. Participants completed eleven map-reading tasks designed to assess different aspects of spatial information processing, including identification, comparison, interpretation, and pattern recognition. The study measured answer accuracy, completion time, and perceived task difficulty. The results indicated that choropleths maps achieved the highest overall accuracy required the shortest response times and were perceived as the easiest map type to use. In contrast, isoline maps generally produced lower accuracy, longer responses times, and higher difficulty ratings. The authors concluded that thematic map type plays a significant role in influencing both objective performance and subjective user experience, and that certain map representations are better suited to particular analytical tasks than others.

Together, these studies provide strong evidence that cartographic representation influences how users interpret and understand spatial information. However, previous research has primarily focused on comparisons involving choropleth, graduated symbol, and isoline. As a result, the relative effectiveness of Dot Density, Filled Isopleths, and Unfilled Isopleths maps for representing spatial distribution patterns under both controlled synthetic and realistic conditions remains unclear. Therefore, this study extends existing cartographic perception research by comparing these thematic map types using objective performance measures (accuracy and response time) as well as subjective evaluations of perceived difficulty.

1.3 Aim of the study

This research aims to fill that gap by testing these three map types using both real-world population data and simulated datasets. It examines how each technique affect user' ability to recognize population patterns such as hotspot and gradient effectively.

1.4 Research Objectives

1.4.1 Main Objective

To assess the impact of different thematic mapping techniques on the interpretation of spatial population distribution patterns.

1.4.2 Specific Objectives

1. To compare the effectiveness of Dot Density maps, Filled Isopleths, and Unfilled Isopleths in hotspot identification tasks.
2. To evaluate how different thematic mapping techniques support spatial gradient interpretation along continuous population profiles.
3. To compare interpretation accuracy and response time across thematic map types.
4. To assess perceived cognitive difficulty associated with hotspot and gradient tasks.
5. To investigate the relationship between subjective task difficulty and objective interpretation performance.
6. To evaluate thematic map usability using both synthetic and real-world population datasets.
7. To provide recommendations for selecting appropriate thematic mapping techniques for population visualization and spatial decision-making tasks.

2. Methods

This study employed a controlled web-based user experiment to investigate how different thematic map representations influence the perception and interpretation of spatial population density patterns. A between-subjects experimental design was adopted, whereby participants were randomly assigned to one of three cartographic conditions: Dot Density (DD), Filled Isopleths (FI), or Unfilled Isopleths (UI). Each participant interacted with only one map type throughout the experiment to minimize learning effects, visual familiarity, and participant fatigue.

The experiment was designed around two common spatial analysis tasks: hotspot identification and spatial gradient interpretation. To evaluate map effectiveness, three performance indicators were measured: interpretation accuracy, response time, and perceived task difficulty. These measures provided complementary insights into the effectiveness, efficiency, and usability of the different thematic representations.

Both real-world and synthetic population datasets were used. Real-world datasets from Germany and the United States provided realistic geographic contexts, while synthetic datasets were generated in Python to create controlled hotspot and gradient patterns. This combination allowed the evaluation of thematic map performance under both realistic and experimentally controlled conditions.

To ensure a fair comparison, all maps were derived from identical underlying datasets and standardized using consistent layouts, scales, legends, and visual design, visual design parameters. Dot Density maps represented population through point symbols, while Filled and Unfilled Isopleths were generated using Kernel Density Estimation (KDE) to create continuous density surfaces and contour-based representations.

Data were collected through an online survey implemented in LimeSurvey and analysed using Chi-square tests, Kruska-Wallis tests, and Generalized Linear Mixed Model (GLMM). These analyses were used to identify differences in interpretation accuracy, response time, and perceived difficulty among the three thematic mapping techniques.

Survey Link: https://survey.plus.ac.at/index.php/624757?lang=en

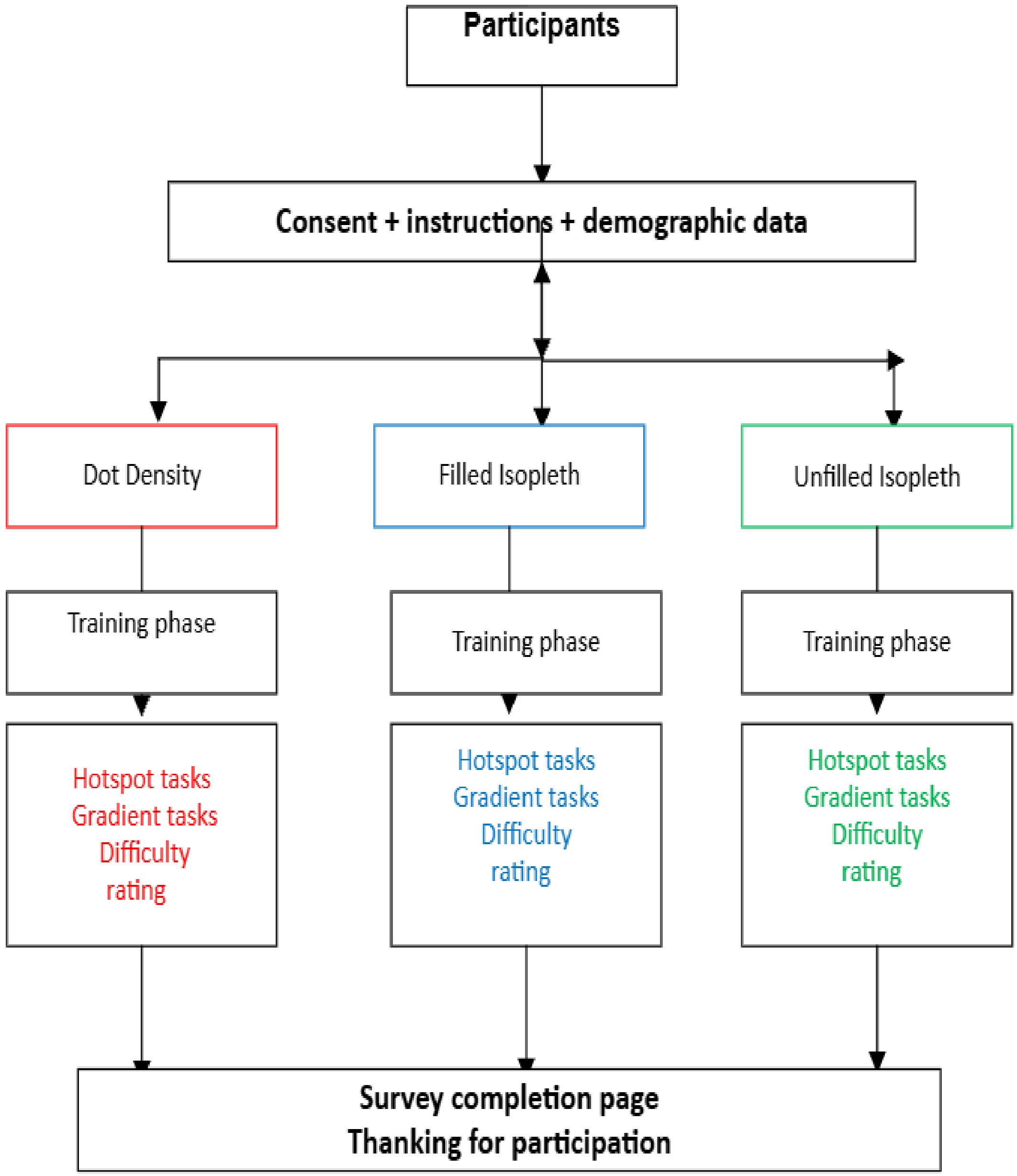

Figure 1: Graphical overview of the experimental workflow and random assignment procedure used on Limesurvey

Data Analysis

Preprocessing and thematic map generation were performed using GIS and spatial analysis software including ArcGIS Pro/Python. The collected data were cleaned on Excel using quantitative statistical methods to evaluate differences in thematic map interpretation performance across the three visualization techniques. Descriptive statistics were used to summarize participants' demographics, interpretation accuracy, response times, and perceived difficulty ratings. Comparative statistical analyses were conducted to evaluate performance differences between Dot Density maps, Filled Isopleths, and Unfilled Isopleths. The analysis also examined relationships between subjective difficulty ratings and objective performance measures. Statistical analyses were conducted using Jamovi a free and open-source statistical software based on R programming language. Jamovi was selected because it combines the computational power and reproducibility of R with a user-friendly graphical interface. The software supports a wide range of statistical procedures, making it suitable for quantitative research and academic applications (Navarro & Foxcroft, 2019; Sahin & Aybek, 2020; Silva de Souza et., 2024; Ashour, 2024)

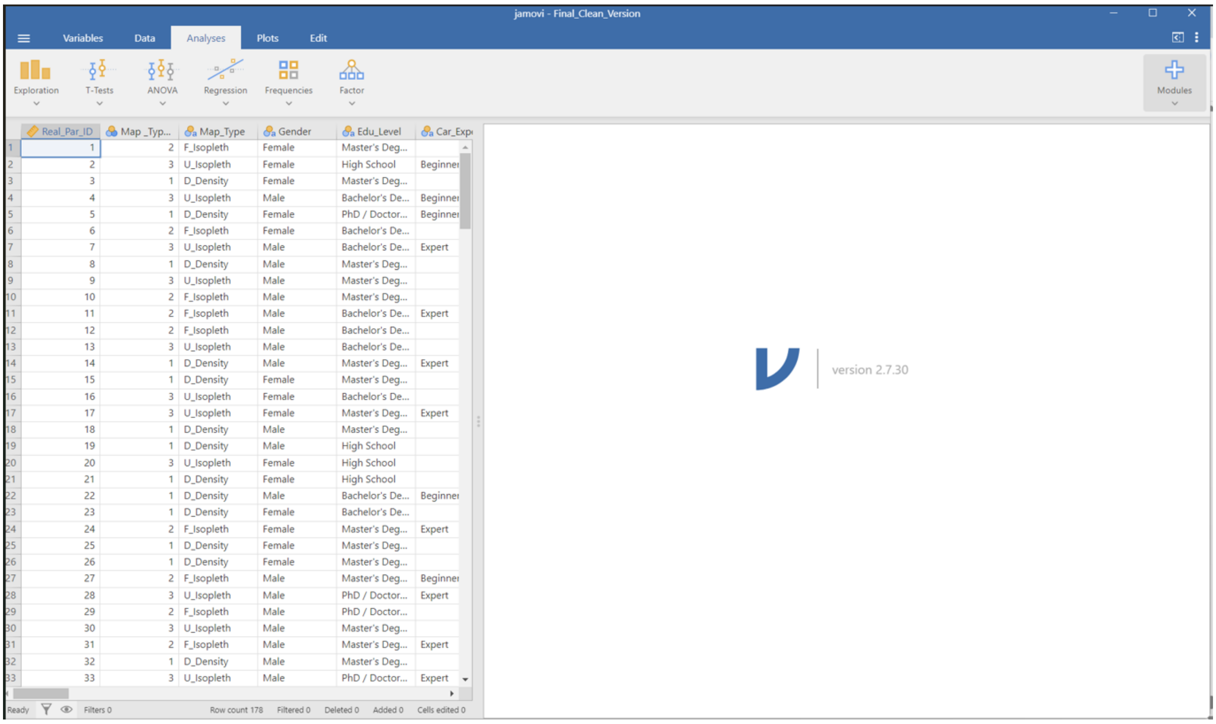

Figure 2: Jamovi (Version 2.7.30) interface displaying the main data view panel

3. Main Thesis Results

The study demonstrated that cartographic representation significantly influences how users perceive and interpret population density patterns. Filled Isopleths achieved the highest overall interpretation accuracy and were generally perceived as the easiest map type to use. Dot Density maps often produced the fastest response times, while Unfilled Isopleths were associated with higher perceived difficulty and lower performance.

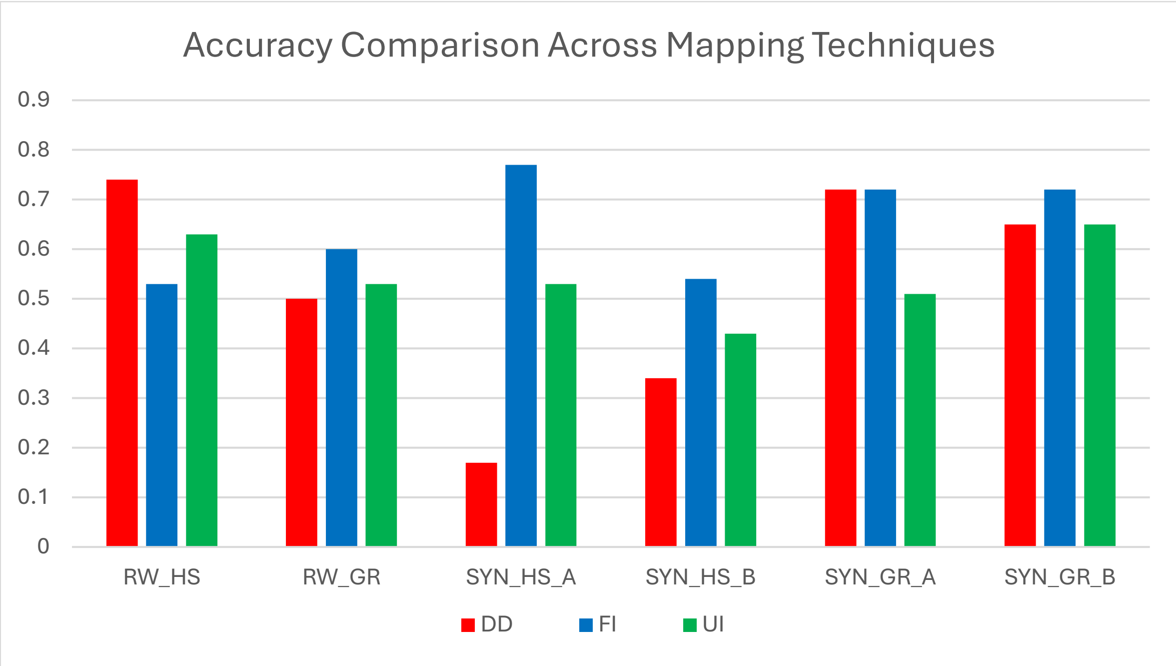

Figure 3: Average accuracy scores across thematic mapping techniques for real-world and synthetic spatial interpretation tasks.

Table 1. Inferential Statistical analysis of answer accuracy across the three thematic map types.

Task	Chi-Square Overall χ^2	*P*	Cramér's V	Pairwise Comparison	χ^2	Cramér's V	*P*	*Significant*
RW_HS	5.66	0.059	0.178	DD-FI	5.64	0.217	0.018	Yes
				DD-UI	1.45	0.112	0.229	---------
				FI-UI	1.38	0.107	0.241	---------
RW_GD	1.17	0.557	0.811	DD-FI	0.13	0.0972	0.136	---------
				DD-UI	0.138	0.0345	0.710	---------
				FI-UI	0.477	0.0628	0.491	---------
SYN_HS_A	43.8	< .001	0.496	DD-FI	43.5	0.602	< .001	Yes
				DD-UI	0.853	0.379	< .001	Yes
				FI-UI	3.09	0.253	0.006	Yes
SYN_HS_B	5.09	0.079	0.169	DD-FI	5.02	0.204	0.025	Yes
				DD-UI	0.908	0.0885	0.341	---------
				FI-UI	1.65	0.117	0.199	---------
SYN_GD_A	7.49	0.024	0.205	DD-FI	0.000	0.002	0.984	---------
				DD-UI	5.27	0.213	0.022	Yes
				FI-UI	5.56	0.215	0.018	Yes
SYN_GD_B	0.926	0.629	0.0712	DD-FI	0.701	0.0764	0.402	---------
				DD-UI	0.00	0.00	1.00	---------
				FI-UI	0.701	0.0764	0.402	---------

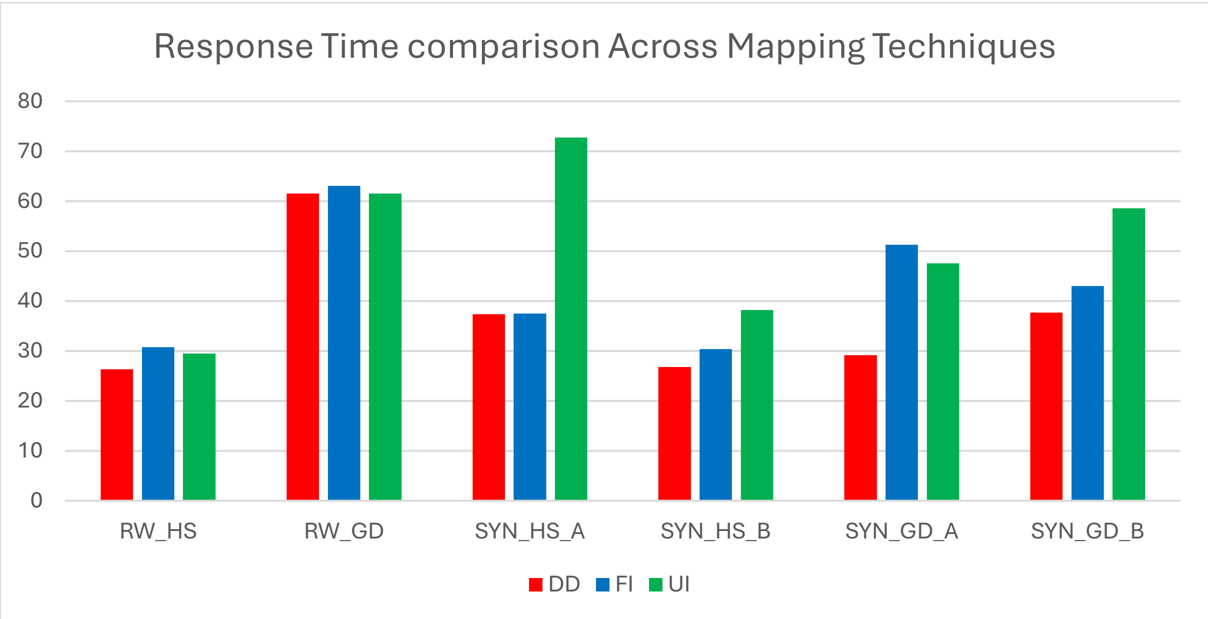

Figure 4: Average response time across thematic mapping techniques for real-world and synthetic spatial interpretation tasks

Table 2. Inferential Statistical analysis of response time across the three thematic map types.

Task	*Kruskal Wallis H*	*P*	Map Type	M (s)	SD	Post Hoc Groups	*p*
RW_HS	2.35	0.309	DD	26.3	15.8	-------	Not Statistically Significant
			FI	30.72	17.9		
			UI	29.5	22.8		
RW_GD	7.00	0.30	DD	61.5	46.0	DD-FI	1.000
			FI	63.0	39.3	DD-UI	0.165
			UI	61.5	81.3	FI-UI	0.33
SYN_GD_A	12.10	0.002	DD	29.2	14.6	DD-FI	0.012
			FI	51.3	77.9	DD-UI	0.005
			UI	47.5	33.7	FI-UI	1.000
SYN_GD_B	4.99	0.082	DD	37.7	25.8	--------	Not Statistically Significant
			FI	43.0	21.2		
			UI	58.6	67,5		
SYN_HS_A	6.67	0.36	DD	37.3	28.9	DD-FI	0.937
			FI	37.5	39.7	DD-UI	0.376
			UI	72.7	137	FI-UI	0.031
SYN_HS_B	2.16	0.340	DD	26.8	19.6		Not statistically Significant
			FI	30.4	18.3		
			UI	38.2	41.7		

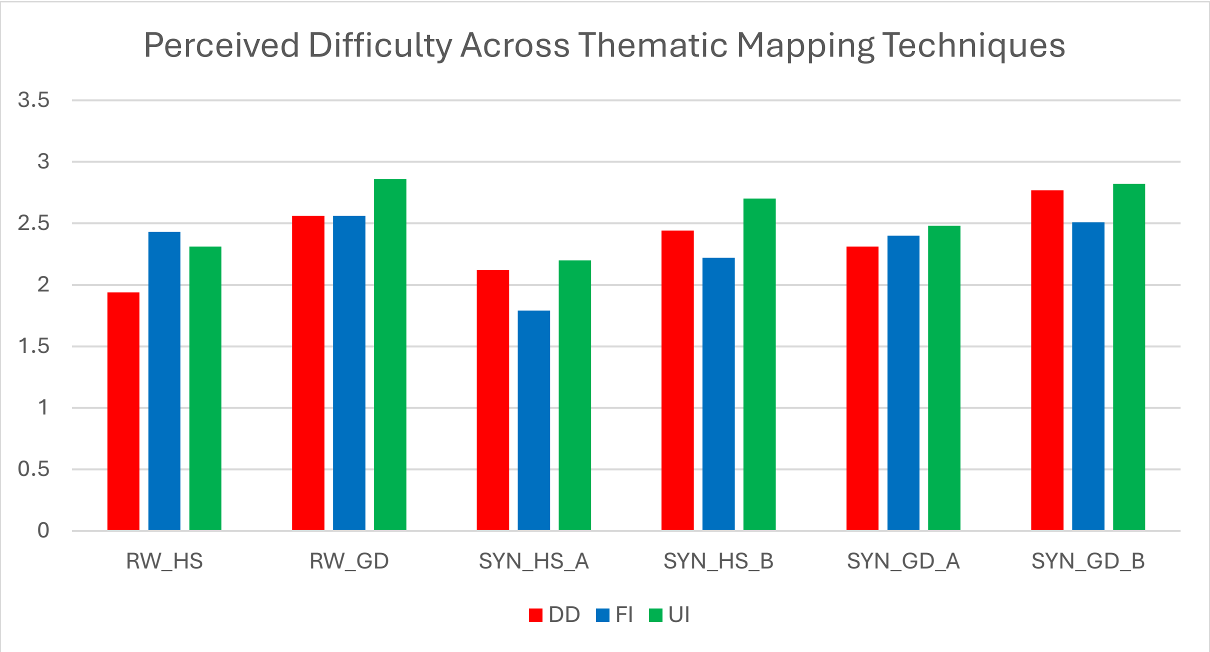

Figure 5: Average difficulty rating across thematic mapping techniques for real-world and synthetic spatial interpretation tasks

Table 3. Inferential Statistical analysis of perceived difficulty rating across the three thematic map types.

Task	**Kruskal-Wallis's X^2**	***p***	**Significant Pairwise Comparison**	***p***
RW_HS	8.37	0.015	DD:FI	0.026
RW_GRD	3.05	0.218	- - - - - - - - - - - -	- - - - - - - - -
SYN_HS_A	7.58	0.023	FI:UI	0.029
SYN_HS_B	7.20	0.027	FI:UI	0.022
SYN_GRD_A	0.79	0.674	- - - - - - - - - - - -	- - - - - - - - -
SYN_GRD_B	4.25	0.119	- - - - - - - - - - - -	- - - - - - - - -

The results further showed that map effectiveness depends on the analytical task being performed. Filled Isopleths were particularly effective for communicating continuous density patterns and supporting spatial gradient interpretation, whereas Dot Density maps remained useful for identifying local clustering and hotspot patterns.

The comparison of synthetic and real-world datasets revealed consistent performance trends across different spatial contexts, suggesting that the findings are applicable to both controlled experimental settings and realistic geographic scenarios.

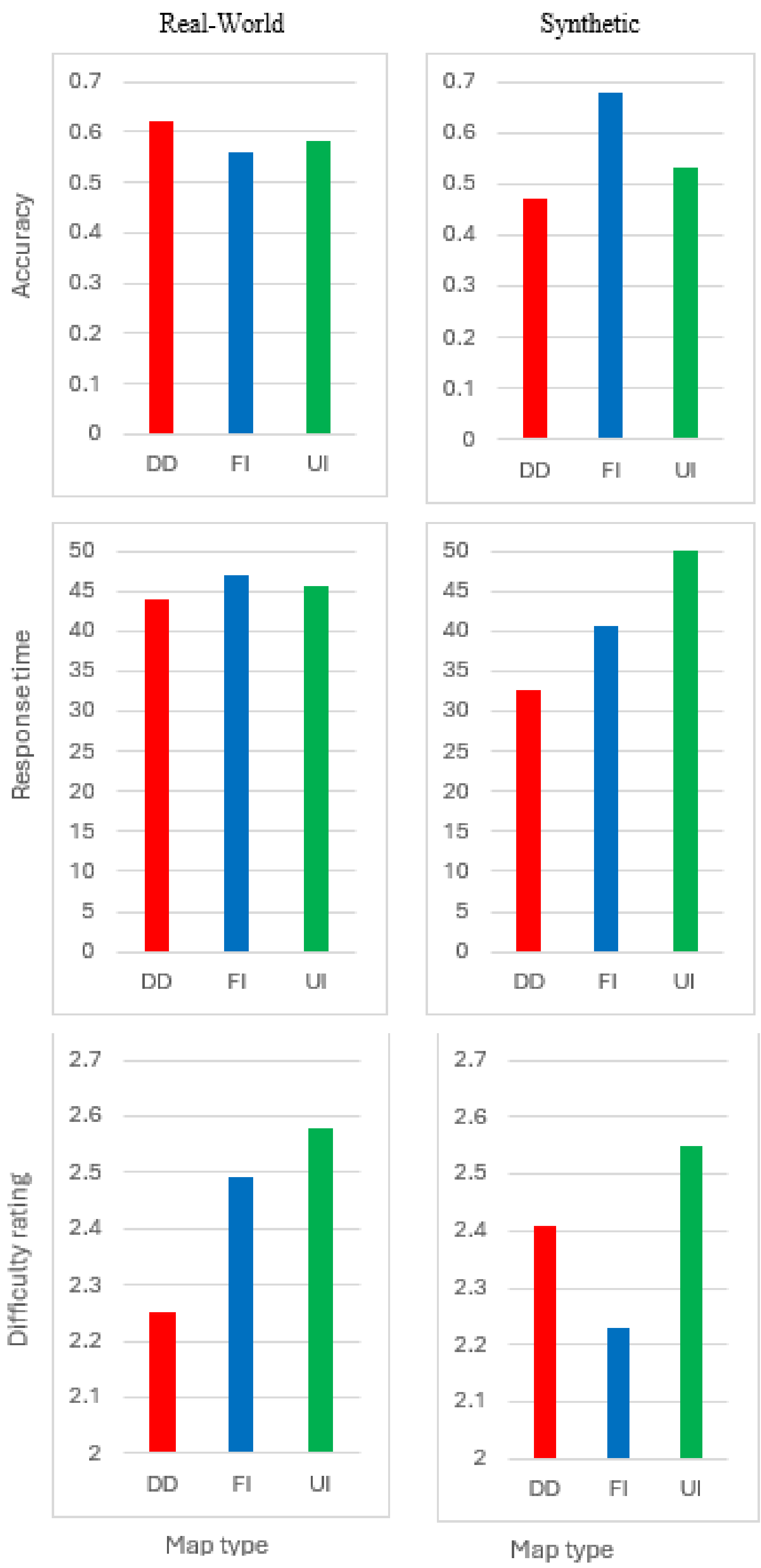

Figure 6: Comparison of interpretation performance between real-world and synthetic spatial tasks across thematic mapping techniques. Real-world tasks include RW_HS and RW_GD, while synthetic tasks include SYN_HS_A, SYN_HS_B, SYN_GD_A, and SYN_GD_B

Additionally, participant characteristics such as age, gender, educational level, level and cartographic experience has little influence on overall performance, indicating that the observed differences were primarily driven by the map representation themselves.

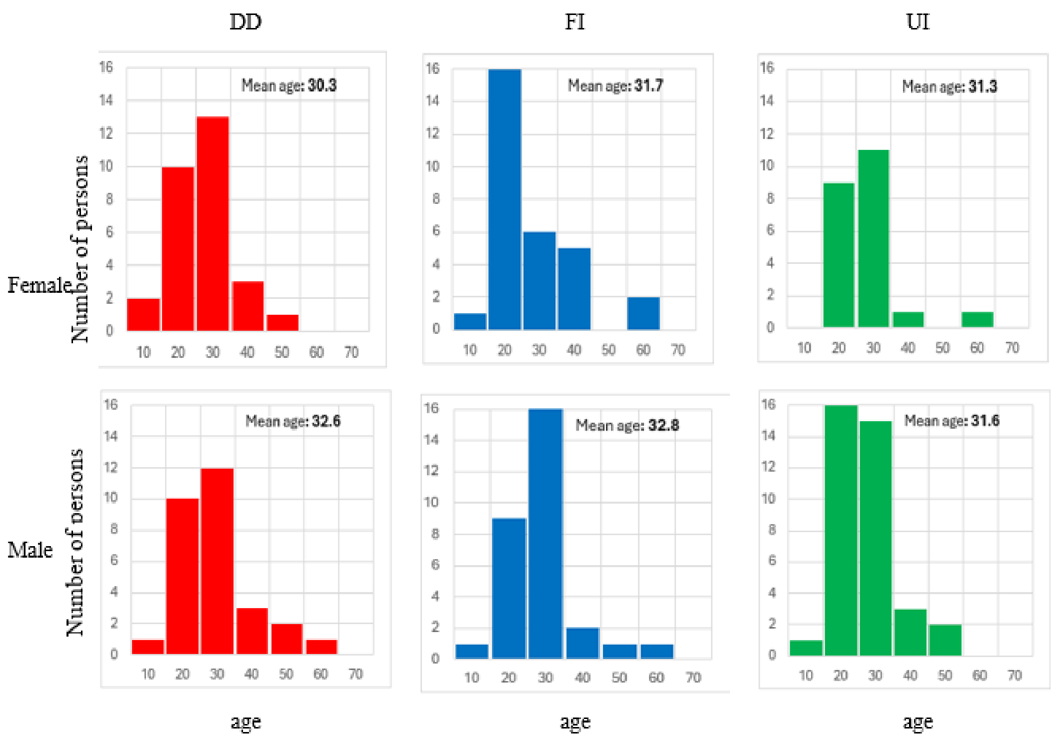

Figure 7: Age distribution and group

Table 1. Participants educational level (Raw count)

	DD	FI	UI	∑
High School	9	10	4	23
Bachelor's Degree	27	31	30	88
Master's Degree	17	15	15	47
PhD / Doctorate	3	4	4	11
Others	2	2	5	9
∑	**58**	**62**	**58**	**178**

Table 2: Self-reported cartographic experience (Raw count)

	DD	FI	UI	∑
Beginner	15	7	9	31
Basic	5	5	5	15
Intermediate	15	23	11	49
Advanced	13	15	24	52
Expert	10	12	9	31
∑	**58**	**62**	**58**	**178**

Overall, the findings highlight the importance of selecting thematic mapping techniques according to the intended analytical task and provide practical recommendations for population density visualization and cartographic design.

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